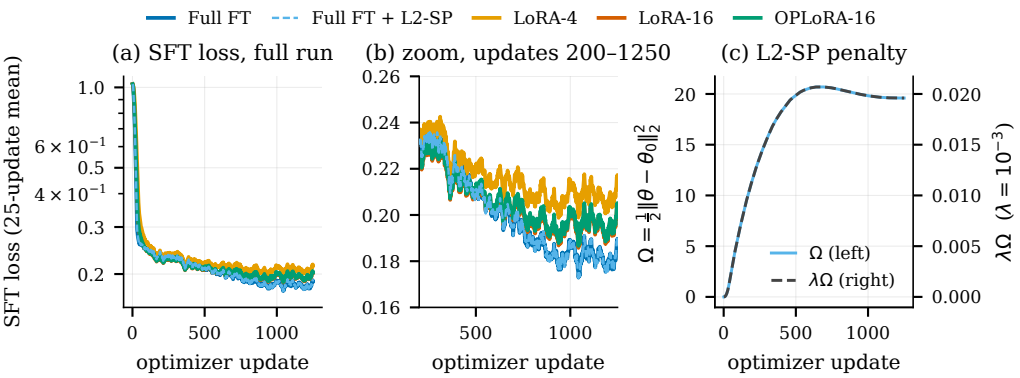

Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(a) SFT loss, full run
(b) zoom, updates 200–1250
(c) L2-SP penalty
SFT loss (25-update mean)
optimizer update
optimizer update
optimizer update
$\Omega = \frac{1}{2}\|\theta - \theta_0\|_2^2$
$\lambda\Omega$ $(\lambda = 10^{-3})$
Ω (left)
λΩ (right)